PHY101: Introduction to Physics-1

Monsoon, 2018

Tutorial-2

1. A particle of mass m is subject to a force $F(t) = \alpha t^3$ and moving along x-direction. The initial position and speed are x_0 and v_0. Find $x(t)$ and $v(t)$. Draw $x(t)$ *vs* t and $v(t)$ *vs* t graphs.

2. Identify the surface for each of the following equations.

(a) $r = 5$

(b) $\theta = \frac{\pi}{3}$

3. Convert the point (-1, 1, $-\sqrt{2}$) from Cartesian to spherical coordinates.

4. Using the expression of infinitesimal volume element expressed in cylindrical polar coordinate system, show that the volume of a cylinder of radius r and height h is $\pi r^2 h$.

5. A special effects studio wants to film a scene in which the Leaning Tower of Pisa (take height, h =100 meters) topples to the ground. In order to achieve this, the studio might make a scale model of the tower, which is (say) 1 meter tall, and then film the model falling over. The only problem is that the resulting footage would look completely unrealistic, because the model tower would fall over too quickly. The studio could easily fix this problem by slowing the film down. The question is by what factor should the film be slowed down in order to make it look realistic? Take m as the mass of the tower and g as the acceleration due to gravity.